MIL-C-14836A(MU)
5 May 1972
SUPERSEDING
MIL-C-14836(MU)
1 August 1968

MILITARY SPECIFICATION

CASE, PROJECTILE, BURSTER, M9A1

1. SCOPE

1.1 This specification covers Case, Projectile, Burster, M9A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified.

SPECIFICATIONS

Federal

O-A-51	- Aceton, Technical
O-C-303	- Chromium Trioxide, Technical (Chromic Acid)
O-O-670	- Orthophosphoric (Phosphoric) Acid, Technical

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For

STANDARDS

Federal

Federal Test Method Standard No. 141 - Paint, Varnish, Lacquer, and Materials; Method of Inspection, Sampling and Testing

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC 1310

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2.1 (Cont'd)

MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering For Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

D10543048	- Case, Projectile, Burster, M9A1
IEL-10543048	- Case, Projectile, Burster, M9A1

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018.)

ASTM Method	- American Society for Testing and Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 916 Race Street, Phila., Pa. 19103.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.1.1 Material.-The supplier shall be responsible for selecting from the classification list contained in the raw material specifications, the surface and physical condition of the raw material intended for use in manufacture.

3.2 Stress relief.-The cold worked Burster Case shall be stress

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3.2 (Cont'd)

relieved at a temperature of 700 degrees Fahrenheit minimum. No work involving plastic flow of metal is permitted after final stress relief.

3.3 Mechanical properties.-Mechanical properties specified on the applicable drawing shall be uniform throughout each burster casing.

3.4 Protective coating.

3.4.1 Prior to application of the phenolic baking varnish, all surfaces of the burster case shall be thoroughly cleaned to remove all traces of grit, oil, wax, grease, rust or other foreign matter.

3.4.2 Final rinse.-The final rinse used in cleaning the burster case prior to application of the phenolic baking varnish, shall be maintained at a ph of 2.0 to 5.0 by the addition of a sufficient amount of chromic acid which shall comply with the requirements of O-C-303 (approximately .08 ounce per gallon of water) or a mixture of phosphoric acid, which shall comply with the requirements of O-O-670, and chromic acid (approximately .04 ounce of each per gallon of water). This rinse shall be maintained at a temperature of 140 degrees F. to 210 degrees F.

3.4.3 Coating application.-The phenolic baking varnish shall be applied to the interior and exterior surfaces of the burster case as required by the applicable drawing, within 4 hours after final cleaning operations have been completed. The varnish applied to the burster case shall produce an adherent film with no visible pinholes, craters, blisters or uneveness. Varnish shall be applied by a spray method unless otherwise authorized by the procuring activity.

3.4.4 Cured coating.-After curing, the varnish coating shall be smooth and continuous and shall not soften or loosen when subjected to the acetone curing test. The acetone shall comply with the requirements of O-A-51.

3.4.5 Salt spray.-The cured varnish coating shall withstand exposure to the 20 percent salt spray for 24 hours without evidence of rust. However, rust within 1/32 inch of all edges shall be permitted.

3.5 Workmanship.-The requirements for workmanship are as shown on the applicable drawing, referenced specifications and the following:

3.5.1 Metal defects.-The burster case shall be free from cracks, splits, cold shuts, inclusions, porosity, laps, seams or other similar defects.

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3.5.2 Burr.-The burster case shall contain no burr which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.-No burster case or assembly shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

3.6 Press fit surface.-The diameter immediately to the rear of the flange is a critical press fit surface and must remain free of dirt and products of corrosion, and must be undamged for its entire length.

3.7 Cleaning prior to testing.-The burster case, prior to hydrostatic and air testing, shall be free from rust and other foreign matter in order not to clog passages and possible result in false non-leakers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary, to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 10 burster cases as follows:

5 Burster casings (coated)
5 Burster casings (uncoated)

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by

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4.2.1 (Cont'd)

the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing processes, procedures, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Burster case lot.-A burster case lot shall consist of burster cases produced by one manufacturer in one unchanged process, in accordance with the same drawings, same drawing revisions, same specification and same specification revisions. The burster case lot thus constituted shall be the basis for selecting the samples for the tension test hereinafter prescribed.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. When performing pressure test, the item under test will be observed for noncompliance for the entire time it is required to be pressurized. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination, except for the press fit area. Surface finish of the press fit area shall be checked after completion of all manufacturing and gaging operations which might affect the surface characteristics. Four power (4X) magnification may be used in performing visual examination.

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4.3.2.1 Case, Projectile, Burster, Drawing D10543048 (prior to varnish).

Defect	Method of Inspection
CRITICAL	
1. Diameter of press fit surface	Gage
2. Presence of reverse taper in press fit surface (max.)	Gage
3. Hydrostatic pressure stamp missing (indicating failure to pass test)	Visual
4. Air pressure stamp missing (indicating failure to pass test)	Visual
5. Metal defective	Visual
MAJOR	
101. Diameter, outside, rear of press fit surface	Gage
102. Length of press fit surface	Gage
103. Thickness of wall	Gage
104. Thickness through base	Gage
MINOR	
201. Diameter of flange	Gage
202. Length from rear of flange to base	Gage
203. Thickness of flange	Gage
204. Radii or chamfers missing or incorrect	Visual (See 6.1)
205. Surface finish improper (except as otherwise classified)	Visual (See 6.1)
206. Burr	Visual
207. Foreign matter	Visual
208. Marking missing, incorrect or illegible	Visual

4.3.2.2 Case, Burster, Drawing D10543048 (after varnish).

CRITICAL	
1. Surface finish of the press fit surface, improper	Measured
MAJOR	
101. Coating inadequate or defective	Visual
102. Corrosion preventive compound missing from press fit area	Visual

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MINOR	
None defined	

4.3.2.3 Carton (prior to sealing).

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Improper assembly	Visual
202. Incorrect number of cases in carton	Visual
203. Separator missing or damaged	Visual

4.3.2.4 Carton (after sealing).

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual and Visual
202. Carton improperly sealed	Manual and Visual
203. Carton damaged, cut or gouged	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical properties (see 6.3).

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4.3.3.1.1 Tension test.-Two burster cases from each lot of 5000 burster cases or fraction thereof shall be subjected to the tension test. Failure of any specimen to comply with the mechanical properties requirements of the applicable drawing shall be cause for rejection of the lot represented by the sample.

4.3.3.2 Hydrostatic test.-Each burster case, prior to gaging and coating, shall be subjected to this test. Failure of any burster case to comply with the requirements of the drawing shall be cause for rejection of the burster case. Burster cases which comply shall be stamped with the letter "H" as indicated on the applicable drawing.

4.3.3.3 Air test.-Each burster case after the hydrostatic test, and prior to coating, shall be subjected to an internal air pressure test to determine compliance with the requirements specified on the applicable drawing. Failure to comply, as evidenced by the presence of bubbles on the surface of the burster case or rising to the surface of the water, shall be cause for rejection of the burster case. Burster cases that comply shall be stamped with the letter "A" as indicated on the applicable drawing.

4.3.3.4 Protective coating.

4.3.3.4.1 Final rinse.-In order to determine the ph of the final rinse, 2 burster casings as they come off the rinse or the rinse itself shall be checked every two hours. Failure of the final rinse to comply with 3.4.2 shall be cause for taking corrective action. Failure to take corrective action shall be cause for suspension of inspection for acceptance until corrective action has been taken.

4.3.3.4.2 Curing test.-One burster case from each 4 hour production or fraction thereof, shall be subjected to this test. Failure of the sample to comply with 3.4.4 shall be cause for rejection of the production represented by the sample.

4.3.3.4.3 Salt spray.-Two burster cases from each production lot shall be subjected to this test. Failure of any burster case to comply with 3.4.5 shall be cause for rejection of the lot.

4.3.4 Straightness of inside diameter.-The inside diameter of each burster case, after protective coating, shall be gaged as prescribed by the applicable drawing. Failure of the functional gage to enter for the full indicated length shall be cause for rejection of the burster case.

4.3.5 Inspection equipment.-Index of Inspection Equipment Lists, IEL-10543048 identifies the inspection equipment required to perform the examination and test prescribed herein. Equipment design shall be in accordance with code designation of the applicable IEL.

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4.4 Test methods.

4.4.1 Mechanical properties (see 6.3).

4.4.1.1 Tension test.-A full section tension test specimen, as prescribed by ASTM Method E-8, with a 1-inch gage length, shall be taken from the tube of each burster case selected in accordance with 4.3.3.1.1. The criteria for determining yield strength and elongation shall be as prescribed by ASTM Method E-8.

4.4.2 Hydrostatic test.-The burster case shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. The burster case shall be mounted in the test fixture and held in place by clamping the sleeve flange; no other portion of the burster case will be restrained during application of pressure. The pressure will be applied for the minimum time specified on the applicable drawing. The equipment will be calibrated prior to the start of each day's operation and at intervals of not over 4 hours during operation. It shall also be calibrated prior to restarting the operation after a stoppage of 2 hours or more.

4.4.3 Air test.-Each burster case shall be mounted in an approved fixture in such a manner that the burster case will be completely submerged in water. When performing this test, the inspector will ascertain that the required pressure has been reached, and the water has been permitted to settle, after immersion of the case under test, prior to start of timing. Observation for leakage shall be made for the entire time the case is required to be pressurized. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours during operation. It shall also be calibrated prior to restarting the operation after a stoppage of 2 hours or more.

4.4.4 Protective coating.-The procedures for testing the cleaning and application of protective coating shall be in accordance with the applicable specifications.

4.4.4.1 Final rinse.-The ph of the final rinse shall be determined by the use of Universal ph papers, or equivalent, by applying the paper to the side of the burster casing immediately after removal from the rinse or the rinse bath itself, and visually examined to determine compliance with the requirements.

4.4.4.2 Curing test.-An area of at least 1 square inch on the sample burster case shall be kept wet with acetone for 2 minutes. The area being tested shall be vigorously rubbed with the thumb and visually inspected for evidence of lifting, blistering, or softening of the varnish coating.

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4.4.4.3 Salt spray test.-Salt spray test shall be performed using equipment and procedures prescribed by Federal Test Method Standard No. 141.

4.4.5 Inside diameter.-The functional gage shall enter into the burster case for the length indicated on the applicable drawing.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level C.-Packing and marking shall be in accordance with MIL-STD-1169.

5.2 Packing.-Burster casings shall be packed in accordance with MIL-STD-1169.

5.3 Marking.-Packaged burster casings shall be marked in accordance with MIL-STD-1169.

5.4 Data card.-The contractor shall prepare and furnish data cards, in the number to be determined by the contracting officer, in accordance with MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination this characteristic may be measured or gaged to determine acceptability.

6.2 Protective coating.

6.2.1 Application.-Care should be taken in the application of the varnish to prevent formation of uncoated areas on the interior due to vapor wash or other causes.

6.2.2 Color.-Slight variations of color of the coated surface due to chromic acid rinse shall not be cause for rejection.

6.2.3 Rinsing.-Adequate rinsing is important in the cleaning process. When the water film on the burster case, after the cleaning operation, maintains its continuity (is free from water break), the surface may be considered free from oil and grease.

6.3 Alternative mechanical properties testing.-Alternative procedures for mechanical properties testing may be used by the contractor when such testing and procedures provide as a minimum the quality assurance specified herein. Prior to applying the alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to or better than the quality assurance provisions specified herein.

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6.4 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU(FA)

Project No. 1310-A213

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